

INDEX

— A —

A1, A0 Signals, 5-2, 5-7, 5-9, 5-21, 9-5
 A15–A13 Signals, 7-6
 A19–A16 Signals, 7-6
 A31–A24 Signals, 4-1, 5-3
 AC Specifications, 10-5
 Access Level, 9-17
 Access Time Calculations, 9-12
 Address Bus, 3-2, 5-3, 5-25
 Address Error Exception, 5-14, 6-6
 Address Registers, 1-4
 Address Space, 2-4, 5-3
 Addressing Modes, 1-8
 Arithmetic/Logical Instruction, 8-30, 8-31
 AS Signal, 3-4, 5-2, 5-3
 Autovector, 5-48
 Autovector Interrupt Acknowledge Cycle, 5-48
 AVEC Signal, 3-5, 5-4, 5-48, 5-53

— B —

BERR Signal, 3-7, 5-4, 5-25, 5-53, 5-55, 6-4
 BG Signal, 3-6, 5-63, 5-66, 5-70
 BGACK Signal, 3-6, 5-62, 5-63, 5-66
 Binary-Coded Decimal, 8-32
 Bit Field Manipulation Instructions, 8-36
 Bit Manipulation Instructions, 8-35
 BKPT Instruction, 5-50
 Flowchart, 6-17
 Block Diagram, 1-2
 BR Signal, 3-6, 5-63, 5-66, 5-70, 5-71
 Breakpoint Acknowledge Cycle, 5-50, 6-17
 Flowchart, 5-50
 Timing, 5-50
 Breakpoint Instruction Exception, 6-17
 Bus, 5-24
 Arbitration, 5-62
 Cycles, 5-1
 Master, 5-1
 Operation, 5-1, 5-24
 Bus Arbitration (MC68020), 5-63

Control Unit, 5-67
 Flowchart, 5-63
 Read-Modify-Write, 5-68
 Timing, 5-63
 Bus Arbitration (MC68EC020), 5-70
 Control Unit, 5-73
 Flowchart, 5-70
 Timing, 5-70
 Two-Wire, 5-75, A-1
 Bus Controller, 5-22, 8-2, 8-5
 Bus Cycles, 5-1, 5-25
 Bus Error Exception, 6-4, 6-21
 Bus Fault, 6-21
 Bus Master, 5-1, 5-25, 5-62
 Bus Operation, 5-24
 Byte Enable Signals, 5-21
 Byte Select Control Signals, 9-5

— C —

Cache, 1-13, 4-1, 5-2, 5-22, 5-62, 8-1, 8-7, 9-11
 Control, 4-3
 Internal Cache Holding Register, 5-21
 Reset, 4-3
 Cache Address Register (CAAR), 1-7, 4-3, 4-4
 Cache Control Register (CACR), 1-7, 4-2, 4-3
 CALLM Instruction, 9-14, 9-16, 9-18
 CAS Instruction, 5-39
 CAS2 Instruction, 5-39
 CDIS Signal, 3-7, 4-3
 CLK Signal, 3-7
 Clock Drivers, 9-10
 Condition Codes, 1-7
 Conditional Branch Instructions, 8-37
 Control Instructions, 8-38
 Coprocessor, 6-25, 7-1
 Classification, 7-4
 Communication Protocol, 7-4
 Conditional Instruction Category, 7-10
 Coprocessor Context Restore Instruction
 Category, 7-22

- Coprocessor Context Save and Context Restore Instruction Categories, 7-16
- Coprocessor Context Save Instruction Category, 7-20
- Coprocessor General Instruction Category, 7-8
- Coprocessor Instructions, 7-1, 7-7
- Format Words, 7-18
- Instruction, 6-25, 7-3
- Instruction Execution, 7-6
- Coprocessor Detected
 - Data-Processing-Related Exceptions, 7-51
 - Exception, 7-49
 - Format Errors, 7-52
 - Illegal Coprocessor Command or Condition Words, 7-51
 - Protocol Violation Exceptions, 7-50
 - System-Related Exceptions, 7-51
- Coprocessor Instruction, 6-25, 7-3
- Coprocessor Interface, 5-53, 7-1, 7-2
- Coprocessor Interface Register (CIR), 7-4, 7-5, 7-6, 7-24
 - Command CIR, 7-25
 - Condition CIR, 7-26
 - Control CIR, 7-24
 - Instruction Address CIR, 7-27
 - Memory Map, 7-24
 - Operand Address CIR, 7-27
 - Operand CIR, 7-26
 - Operation Word CIR, 7-25
 - Register Select CIR, 7-27
 - Response CIR, 7-24
 - Restore CIR, 7-25
 - Save CIR, 7-25
 - Selection, 7-6
- CPU Address Space, 2-4, 5-44, 7-6
- CPU Space Type, 5-44, 5-53
- Cycle
 - Asynchronous Cycles, 5-1, 5-5
 - Autovector Interrupt Acknowledge Cycle, 5-45, 5-48
 - Breakpoint Acknowledge Cycle, 5-50
 - Coprocessor Interface Bus Cycles, 7-4
 - Interrupt Acknowledge Cycle, 5-4, 5-45
 - Operand Transfer Cycle, 5-5
 - Synchronous Cycle, 5-24

— D —

- Data Accesses, 4-2
- Data Bus (D31–D0), 3-2, 5-3, 5-5, 5-21, 5-25
- Data Registers, 1-4
- Data Types, 1-8
- DBEN Signal, 3-5, 5-4
- DC Electrical Characteristics
 - MC68020, 10-4
 - MC68EC020, 10-5
- Destination Function Code Register (DFC), 1-7
- Differences between MC68020 and MC68EC020, 1-1, 5-62
- Double Bus Fault, 5-60
- DS Signal, 3-4, 5-4, 5-21
- DSACK1, DSACK0 Signals, 3-5, 5-4, 5-5, 5-24, 5-46, 5-53, 9-5
- Dynamic Bus Sizing, 5-5

— E —

- ECS Signal, 3-4, 5-3
- Effective Address, 8-13, 8-14, 8-16, 8-17, 8-19
- Electrical Specifications, 10-1
- Exception, 2-5, 7-57
 - Address Error Exception, 5-14, 6-6, 7-57
 - Breakpoint Instruction, 6-17
 - Bus Error Exception, 6-4, 6-21
 - Coprocessor-Detected Exception, 7-49
 - cpTRAPcc Instruction Traps, 7-55
 - Data-Processing-Related Exception, 7-51
 - F-Line Emulator Exception, 7-54
 - Format Error Exception, 6-10, 7-57
 - Illegal Instruction, 6-7
 - Interrupt Exception, 5-45, 6-11, 7-56
 - Multiple, 6-17
 - Privilege Violation Exception, 6-7, 6-8, 7-55
 - Protocol Violation, 7-50
 - Reset Exception, 6-4
 - Stack Frames, 6-25
 - System-Related Exception, 7-51
 - Trace Exception, 6-9, 7-55
 - Trap Exception, 6-6
 - Unimplemented Instruction, 6-7
- Exception Handler, 6-2
- Exception Processing, 2-1, 2-5, 6-1
- Exception-Related Instructions, 8-39
- Exception Stack Frame, 2-6, 6-25
- Exception Vector Table, 2-5, 6-2

— F —

FC2–FC0 Signals, 2-2, 2-4, 3-2, 5-2, 5-3, 5-44, 7-6

Features, 1-2

F-Line Operation Words, 7-3

Floating-Point Coprocessor, 7-1, 9-1

Flowchart

- Breakpoint Acknowledge Cycle, 5-50
- Byte Read Cycle, 5-26
- Interrupt Acknowledge Cycle, 5-46
- Long-Word Read Cycle, 5-26
- MC68EC020 Bus Arbitration, 5-70
- MC68020 Bus Arbitration, 5-63
- Read-Modify-Write Cycle, 5-39
- Reset Exception, 6-4
- Write Cycle, 5-33

Format Error Exception, 6-10

— G —

Ground Connections, 3-7, 9-9

— H —

HALT Signal, 3-7, 5-4, 5-25, 5-53, 5-60

— I —

Idle Clock Cycles, 8-7

Illegal Instruction Exception, 6-7

Instruction

- Arithmetic/Logical, 8-30, 8-31
- Bcc, 8-37
- Bit Field Manipulation, 8-36
- Bit Manipulation, 8-35
- BKPT, 5-50
- CALLM, 9-14, 9-16, 9-18
- CAS, 5-39
- CAS2, 5-39
- Control, 8-38
- Coprocessor Conditional Instructions, 7-10
- Coprocessor Context Restore Instruction Category, 7-22
- Coprocessor Context Save Instruction Category, 7-20
- Coprocessor Instruction, 6-25, 7-1, 7-3, 7-7
- Coprocessor Instruction Execution, 7-6

- Coprocessor Context Save and Context Restore Instruction Categories, 7-16
- Coprocessor General Instruction Category, 7-8
- cpBcc, 7-12
- cpDBcc, 7-14
- cpRESTORE, 7-17, 7-22
- cpSAVE, 7-17, 7-20
- cpScc, 7-13
- cpTRAPcc, 7-15, 7-55
- Exception-Related, 8-39
- Illegal Instruction, 6-7
- MOVE, 8-20
- MOVE SR, 8-3
- MOVEA, 8-20
- MOVEC, 4-3
- NOP, 5-62, 8-3
- Prefetches, 4-1
- Primitive Instructions, 7-27
- RESET, 5-76, 7-58
- RTE, 6-19, 6-24
- RTM, 9-14, 9-16, 9-19
- Shift/Rotate, 8-34
- Single-Operand Instructions, 8-33
- Special-Purpose MOVE, 8-29
- STOP, 6-10
- TAS, 5-39
- Unimplemented Instruction, 6-7

Instruction Execution, 5-62, 8-1

Instruction Execution Overlap, 8-4

Instruction Pipe, 1-12, 4-1, 6-21

Instruction Prefetches, 8-1

Instruction Set, 1-10

Instruction Timing, 8-8, 8-9

Internal Cache Holding Register, 5-21

Interrupt, 6-1

- Flowchart, 6-14
- Interrupt Exception, 6-11
- Nonmaskable, 6-12

Interrupt Acknowledge Cycle, 5-4, 5-45, 6-16

- Timing, 5-46

Interrupt Exception, 5-45, 6-11

Interrupt Priority Mask, 5-45, 6-11

Interrupt Stack Pointer (ISP), 1-4, 2-2

IPEND Signal, 3-5, 6-14

IPL2–IPL0 Signals 3-5, 5-45, 6-11

— L —

Long-Word Operand, 5-10, 5-14
Long-Word Read Cycle, 5-26
Long-Word Write Cycle, 5-33

— M —

M-Bit (SR), 1-7, 2-2
Main Processor Detected
 Address Error, 7-57
 Bus Faults, 7-57
 cpTRAPcc Instruction Traps, 7-55
 Exceptions, 7-52
 F-Line Emulator Exception, 7-54
 Format Error, 7-57
 Interrupts, 7-56
 Privilege Violations, 7-55
 Protocol Violation, 7-52
 Trace Exception, 7-55
Master Stack Pointer (MSP), 1-4, 2-2
Maximum Ratings, 10-1
MC68881/MC68882 Floating-Point Coprocessors,
 9-1
Memory Interface, 9-11
Misaligned
 Operand, 5-6, 5-14, 8-2
 Transfer, 5-1, 5-5
Module, 9-14
Module Call, 9-18
Module Return, 9-19
Module Stack Frame, 9-16
MOVE Instruction, 8-20
MOVE SR Instruction, 8-3
MOVEA Instruction, 8-20
MOVEC Instruction, 4-3

— N —

Nonmaskable Interrupt, 6-12
NOP Instruction, 5-62, 8-3
Normal Processing State, 2-1

— O —

OCS Signal, 3-4, 5-3
Ordering Information
 MC68020, 11-1
 MC68EC020, 11-1
Overlap, 8-3

— P —

Package Dimensions
 MC68020 FC Suffix, 11-6
 MC68020 FE Suffix, 11-7
 MC68020 RC Suffix, 11-3
 MC68020 RP Suffix, 11-4
 MC68EC020 FG Suffix, 11-11
 MC68EC020 RP Suffix, 11-9
Pin Assignment
 MC68020 FC Suffix, 11-5
 MC68020 FE Suffix, 11-5
 MC68020 RC Suffix, 11-2
 MC68020 RP Suffix, 11-2
 MC68EC020 FG Suffix, 11-10
 MC68EC020 RP Suffix, 11-8
Port Size, 5-1, 5-5, 5-21, 9-5
Power Supply, 3-7, 9-9
Primitive, 7-4, 7-27
 Busy Response Primitive, 7-30
 CA Bit, 7-29
 DR Bit, 7-29
 Evaluate and Transfer Effective Address
 Primitive, 7-35
 Evaluate Effective Address and Transfer Data
 Primitive, 7-35
 Format, 7-28
 Null Coprocessor Response Primitive, 7-31
 PC Bit, 7-29
 Supervisor Check Primitive, 7-33
 Take Address and Transfer Data Primitive,
 7-39
 Take Midinstruction Exception Primitive, 7-47
 Take Postinstruction Exception Primitive,
 7-48
 Take Preinstruction Exception Primitive, 7-45
 Transfer from Instruction Stream Primitive,
 7-34
 Transfer Main Processor Control Register
 Primitive, 7-41
 Transfer Multiple Coprocessor Registers
 Primitive, 7-42
 Transfer Multiple Main Processor Registers
 Primitive, 7-42
 Transfer Operation Word Primitive, 7-33
 Transfer Single Main Processor Register
 Primitive, 7-40

Transfer Status Register and the scanPC
 Primitive, 7-44
 Transfer to/from Top of Stack Primitive, 7-40
 Write to Previously Evaluated Effective
 Address Primitive, 7-37
 Privilege Level, 2-2
 Changing, 2-3
 Supervisor Level, 1-4, 2-2
 User Level, 1-4, 2-2
 Privilege Violation Exception, 6-7, 6-8
 Processing States, 2-1
 Program Counter (PC), 1-4
 Programming Model, 1-4, 7-1, 7-2

— R —

Read Cycle, 5-3, 5-4, 5-8, 5-14, 5-16, 5-18, 5-22,
 5-26
 Byte Read Cycle, 5-26
 Long-Word Read Cycle, 5-26, 8-2
 Timing, 5-26
 Read-Modify-Write Cycle, 5-3, 5-39, 5-42
 Timing, 5-39
 Registers
 Address Registers, 1-4
 CAAR, 1-7, 4-3, 4-4
 CACR, 1-7, 4-2, 4-3
 Data Registers, 1-4
 DFC, 1-7
 Internal Cache Holding Register, 5-21
 Program Counter (PC), 1-4
 SFC, 1-7
 SR, 1-7, 4-1
 VBR, 1-7
 Reset, 4-3
 Flowchart, 6-4
 Reset Exception, 6-4
 RESET Instruction, 7-58
 RESET Signal, 3-6, 5-76, 6-4
 Reset Exception, 6-4
 RESET Instruction, 5-76
 RESET Signal, 3-6, 5-76, 6-4
 Retry, 5-56
 RMC Signal, 3-4, 5-3, 5-39
 RTE Instruction, 6-19, 6-24
 RTM Instruction, 9-14, 9-16, 9-19
 R/W Signal, 3-4, 5-2, 5-3, 9-5

— S —

S-bit (SR), 1-7, 2-2, 2-3
 Save and Restore Operations, 8-40
 scanPC, 7-28
 Sequencer, 8-2, 8-5
 Shift/Rotate Instructions, 8-34
 Signal(s), 3-8
 A1,A0, 5-2, 5-7, 5-9, 5-21, 9-5
 A15–A13, 7-6
 A19–A16, 7-6
 A31–A24, 4-1, 5-3
 Address Bus, 3-2, 5-3
 AS, 3-4, 5-2, 5-3
 AVEC, 3-5, 5-4, 5-48, 5-53
 BERR, 3-7, 5-4, 5-25, 5-53, 5-55, 6-4
 BG, 3-6, 5-63, 5-66, 5-70, 5-71
 BGACK, 3-6, 5-62, 5-63, 5-66
 BR, 3-6, 5-63, 5-66, 5-70
 Byte Select Control Signals, 9-5
 CDIS, 3-7, 4-3
 CLK, 3-7
 D31–D0, 3-2, 5-3
 DBEN, 3-5, 5-4
 DS, 3-4, 5-4, 5-21
 DSACK1, DSACK0, 3-5, 5-4, 5-5, 5-24, 5-46,
 5-53, 9-5
 ECS, 3-4, 5-3
 FC2–FC0, 2-4, 3-2, 5-2, 5-3, 5-44, 7-6
 Functional Groups, 3-1
 HALT, 3-7, 5-4, 5-25, 5-53, 5-60
 Input Signal, 5-2
 Internal Signal, 5-2
 IPEND, 3-5, 6-14
 IPL2–IPL0, 3-5, 6-11
 OCS, 3-4, 5-3
 RESET, 3-6, 5-76, 6-4
 RMC, 3-4, 5-3, 5-39
 R/W, 3-4, 5-2, 5-3, 9-5
 SIZ1, SIZ0, 3-2, 5-2, 5-3, 5-7, 5-9, 5-21, 9-5
 Single-Operand Instruction, 8-33
 SIZ1, SIZ0 Signals, 3-2, 5-2, 5-3, 5-7, 5-9,
 5-21, 9-5
 Source Function Code Register (SFC), 1-7
 Special-Purpose MOVE Instruction, 8-29
 Special Status Word (SSW), 6-21
 Spurious Interrupt, 5-48

Stack Frame	Transfer, 5-10, 5-14, 5-25
Midinstruction, 7-47	Bus Transfer, 5-1
Postinstruction, 7-48	Direction, 5-3
Preinstruction, 7-46	Misaligned, 5-1, 5-5
Status Register (SR), 1-7, 4-1, 5-45, 6-1	Operand Transfer, 5-1, 5-5
STOP Instruction, 6-10	Trap Exception, 6-6
Supervisor Privilege Level, 1-4, 2-2	— U —
Supervisor Stack Pointer (SSP), 1-4, 2-2	Unimplemented Instruction (F-Line Opcode)
Synchronous Cycles, 5-24	Exception, 6-7
— T —	User Privilege Level, 1-4, 2-2
T1, T0 Bits (SR), 1-7, 6-9	User Stack Pointer (USP), 1-4, 2-2
TAS Instruction, 5-39	— V —
Thermal Characteristics, 10-1	V _{CC} Connections, 3-7, 9-9
MC68020, 10-2	Vector Base Register (VBR), 1-7, 2-5, 6-2
MC68020 CQFP Package, 10-2	Virtual Machine, 1-12
MC68EC020, 10-4	Virtual Memory, 1-10
MC68EC020 PQFP Package, 10-4	— W —
Thermal Resistance, 10-2, 10-4	Write Cycle, 5-3, 5-9, 5-10, 5-12, 5-14, 5-16,
Timing, 5-26, 5-33	5-18, 5-22, 5-33, 5-38
Trace Exception, 6-9	Long-Word Write Cycle, 5-33
Trace Modes, 1-7	Timing, 5-33
Tracing, 6-9	